| Ex:No: 1F<br>Date: 28-01-26 | Household Electricity Consumption Analysis Using Matrix |
|---|---|

**Problem Description:**

An electricity board records monthly electricity consumption (in units) of multiple households. The data is stored in a matrix where **rows represent households** and **columns represent months**. Write an R program to analyze electricity usage by calculating row-wise total consumption, column-wise average consumption, highest usage, and lowest usage.

**Objective:**

To demonstrate the use of matrices in R for storing and analyzing two-dimensional numerical data such as electricity consumption records.

| Sample Input | Expected Output |
|---|---|
| Enter number of households: 3<br>Enter number of months: 3<br>Enter electricity units:<br>Household 1 Month 1 : 250<br>Household 1 Month 2 : 225<br>Household 1 Month 3 : 150<br>Household 2 Month 1 : 300<br>Household 2 Month 2 : 321<br>Household 2 Month 3 : 212<br>Household 3 Month 1 : 112<br>Household 3 Month 2 : 142<br>Household 3 Month 3 : 155 | Electricity Consumption Matrix:<br>Jan Feb Mar<br>Household1 250 225 150<br>Household2 300 321 212<br>Household3 112 142 155<br>Total Consumption per Household:<br>Household1 Household2 Household3<br>625 833 409<br>Average Consumption per Month:<br>Jan Feb Mar<br>220.6667 229.3333 172.3333<br>Highest Consumption: 321<br>Lowest Consumption: 112 |

**Program:**

```
# Electricity Consumption Analysis using Matrix with Row and Column Names
# User input households, months and units
h<-as.integer(readline(prompt="Enter number of
                                households: "))
m<-as.integer(readline(prompt="Enter number of months:
units<-matrix(0, nrow=h, ncol=m)
for(i in 1:h){
  for(j in 1:m){
    units[i,j]<-as.numeric(
      readline(prompt=paste("Household", i, "Month",
                                j, ": ")))
}
}
```

SAMPLE INPUT:

Enter number of households: 2
Enter number of months: 3
Enter electricity units:

Household 1 Month 1: 120
Household 1 Month 2: 140
Household 1 Month 3: 160

Household 2 Month 1: 180
Household 2 Month 2: 200
Household 2 Month 3: 220

EXPECTED OUTPUT:

Electricity Consumption Matrix:

| | Jan | Feb | Mar |
|---|---|---|---|
| Household 1 | 120 | 140 | 160 |
| Household 2 | 180 | 200 | 220 |

Total Consumption per Household:

| Household 1: | Household 2 |
|---|---|
| 420 | 600 |

Average Consumption per Month:

| Jan | Feb | Mar |
|---|---|---|
| 150 | 170 | 190 |

Highest Consumption: 220
Lowest Consumption: 120

# Assign row and column names

```
rownames(units) <- paste("Household", 1:h)
colnames(units) <- paste("Month", 1:m)
```

# Display matrix

```
cat("Electricity Consumption Matrix:\n")
print(units)
```

# Row-wise total

```
row_total <- rowSums(units)
cat("\n Row-wise Total Consumption:\n")
print(row_total)
```

# Column-wise average

```
col_avg <- colMeans(units)
cat("\n Column-wise Average Consumption:\n")
print(col_avg)
```

# Maximum and minimum consumption

```
max_usage <- max(units)
min_usage <- min(units)
cat("\n Highest Consumption: ", max_usage, "\n")
cat("Lowest Consumption: ", min_usage, "\n")
```

**Result:**

Thus, the program analyzes household electricity consumption using matrix operations in R. It demonstrates efficient processing of two-dimensional data through row-wise and column-wise computations.